

Quatro Dispatch

Hourly ride demand for every Addis Ababa zone, two weeks ahead.

**Operations must staff 12 zones across 336 hours
with no view of demand.**

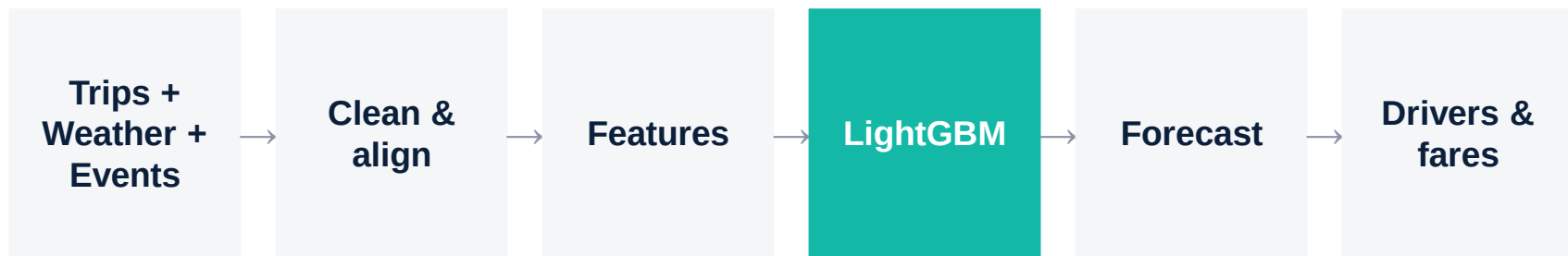
4,032

zone-hours to forecast
 $12 \text{ zones} \times 14 \text{ days} \times 24 \text{ hours}$

1.3 trips
per driver-hour

250 ETB
gross fare per trip

One pipeline turns three messy data streams into a staffing plan.



Exposed through an interactive Streamlit app.

We fixed the data before we modeled it.

11:00

temperature peak, raw UTC



14:00

peak after +3h shift to local time

85,460 → 83,104

rows after removing
duplicates

8 aliases → 12

canonical zones

6 of 6

integrity checks pass

Four measurable forces drive demand, and each became a feature.

+32% / -50%

Weekends: Bole up, admin hubs down

Feature: zone × weekday

+14–24%

Rain uplift [VERIFY: controlled?]

Feature: rain intensity

+52%

Football post-match surge

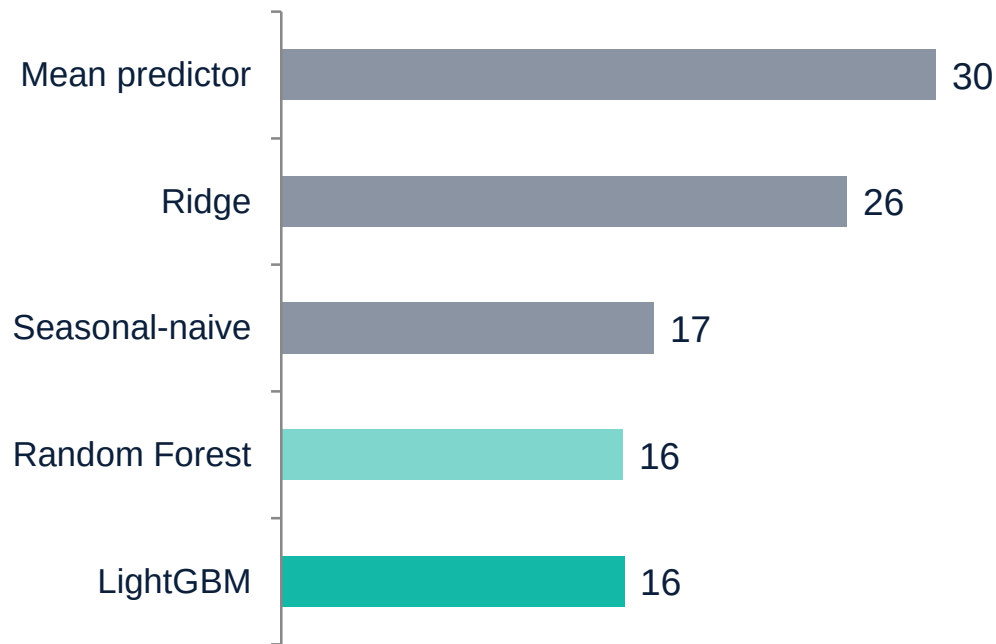
Feature: pre / during / post windows

+55.3%

Demand growth, Jan to Oct

Feature: days_elapsed trend

LightGBM matches Random Forest's accuracy at 8× the speed.



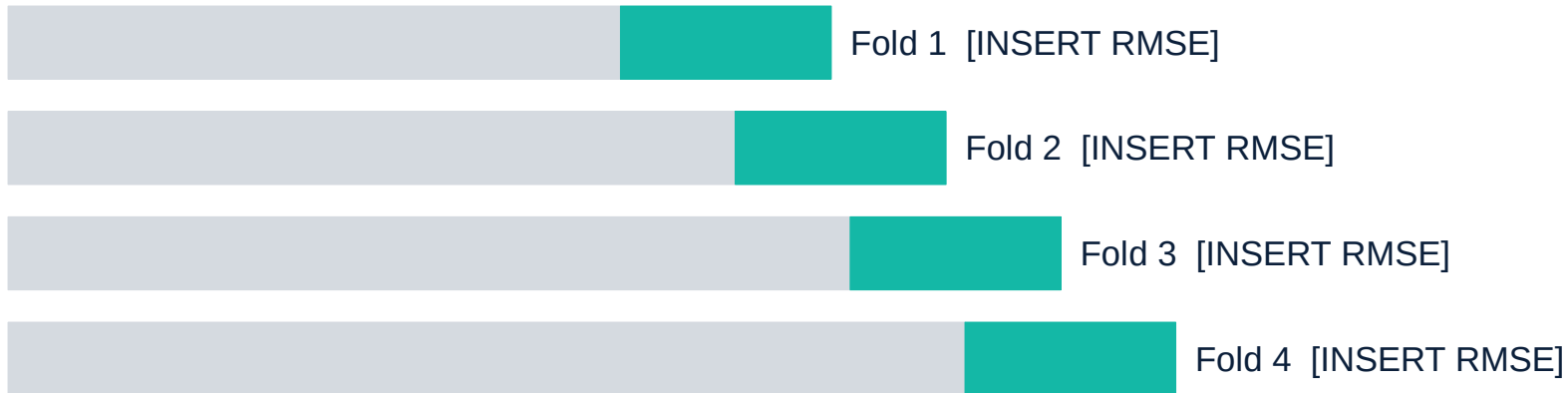
RMSE, lower is better

3s vs 24s

training time, LightGBM vs
Random Forest

Native handling of the 12
zones

Every score is measured on the future of its training data.



Train (always earlier) ▪ Test: 14 days (teal) | Mean RMSE 13.97 ± 2.58

Time-ordered splits only

**avg_fare_birr, avg_wait_min,
active_drivers dropped**

**Features known at forecast
time [VERIFY]**

MAE of 6.8 trips per zone-hour, 10% below the seasonal baseline.

6.8

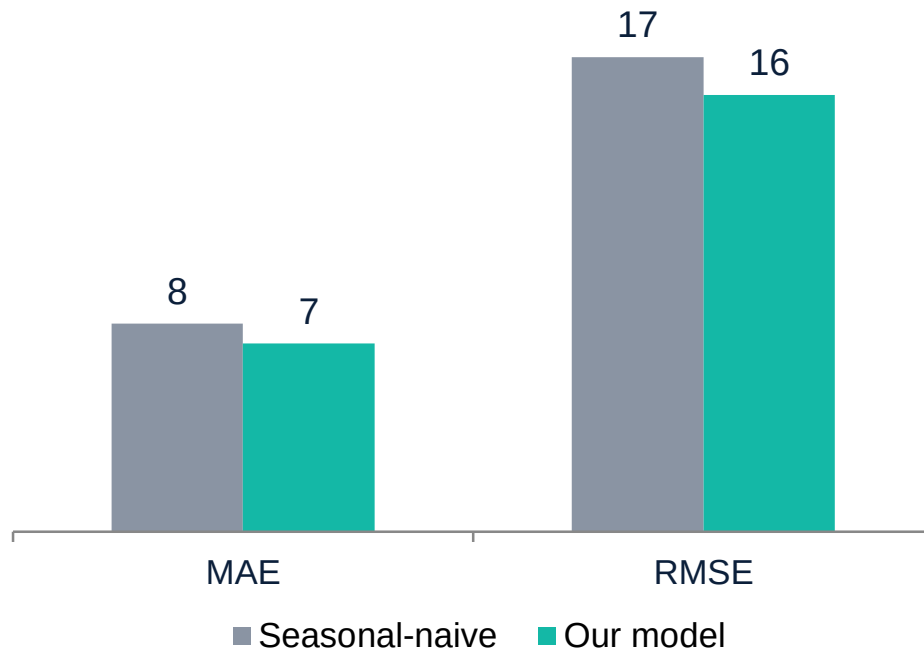
MAE, trips per zone-hour (20% of 33.4 mean)

15.8

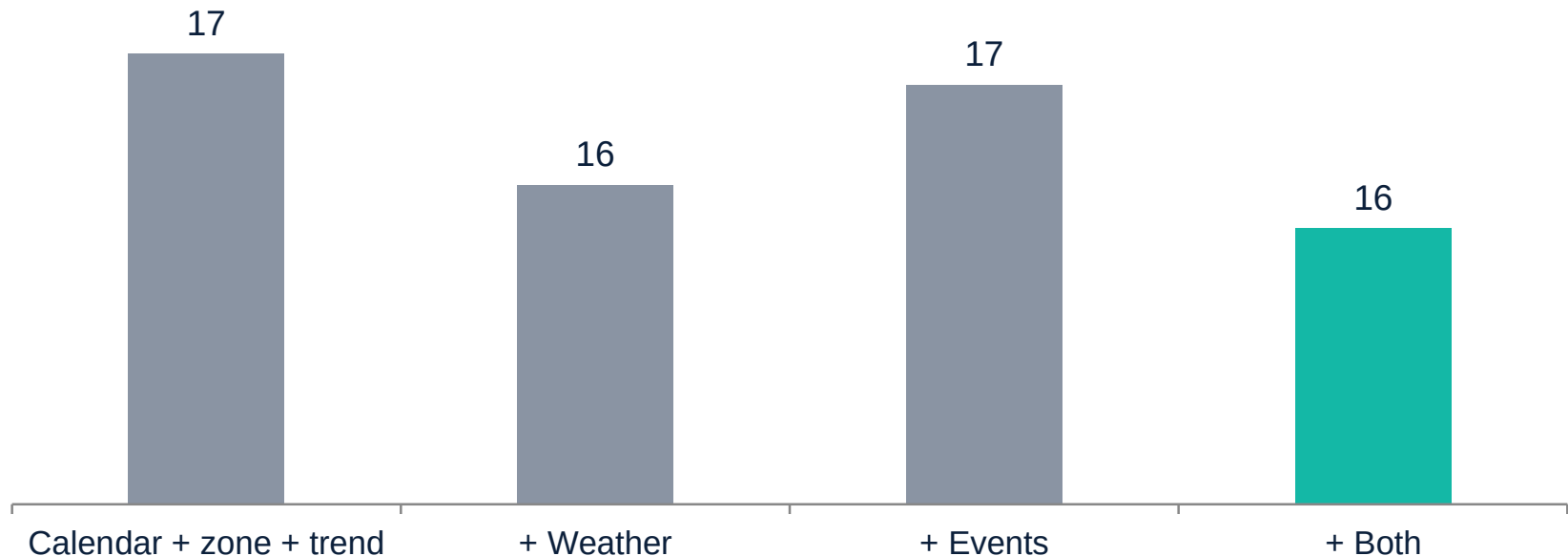
RMSE (baseline 17.16, -8%)

4,032

predictions, no nulls or negatives



Weather and events together cut RMSE from 16.92 to 15.79.



RMSE, lower is better. Axis starts at 14.

A dispatcher picks a zone and date and gets the staffing answer.

[INSERT APP SCREENSHOT]

Peak hour

Drivers needed

Gross fares

Typical error: MAE 6.8 trips $\approx \pm 5$ drivers per zone-hour. RMSE buffer ≈ 12 drivers.
[INSERT live example]

The model misses what the calendar does not know.

Limitations

- Under-forecasts rare spikes (CMC Oct 7: 592 trips, cause unverified)
- Unlisted events are invisible
- About ten months of history
- Tree trend cannot extrapolate

Next steps

- Event feed integration
- Asymmetric loss for under-forecasts (to be tested)
- Uncertainty ranges for staffing buffers

Quatro Dispatch

Clean data, honest validation, a working tool.

Give us any zone and any date from Nov 1–14.

Data · Analysis · Modeling · Deployment